

Causal survival estimands under the AFT data-fusion model with HDP error

Working notes

May 11, 2026

1 Setup

We work under the AFT decomposition

$$\log T_i = m_0(X_i, S_i) + A_i \tau(X_i) + (1 - S_i) A_i c(X_i) + \sigma \varepsilon_i, \quad \varepsilon_i \sim P, \quad (1)$$

with P assigned a hierarchical Dirichlet process (HDP) prior with Gaussian kernels and atoms shared across the two data sources. Conditional on the sampler state, the implied error CDF in source s is

$$\Phi_s(z) = \sum_{k=1}^K \pi_{s,k} \Phi^N\left(\frac{z - \mu_k}{\tau_k}\right), \quad (2)$$

where Φ^N is the standard normal CDF, $\{(\mu_k, \tau_k)\}_{k=1}^K$ are the cluster atoms (shared across sources via the top-level DP), and $\pi_{s,k}$ are the source-specific weights.

The treatment-specific AFT shift at target population s_t is

$$\eta_a(x, s_t) := m_0(x, s_t) + a \tau(x) + (1 - s_t) a c(x), \quad a \in \{0, 1\}. \quad (3)$$

The potential survival function is

$$S_a(t \mid x, s_t) = 1 - \Phi_{s_t}\left(\frac{\log t - \eta_a(x, s_t)}{\sigma}\right), \quad (4)$$

and we collect causal estimands as functionals of $\{S_0(\cdot \mid x, s_t), S_1(\cdot \mid x, s_t)\}$.

For the closed-form expressions below it is convenient to define, for each component k of the HDP mixture,

$$\xi_{a,k}(x, s_t) := \eta_a(x, s_t) + \sigma \mu_k, \quad \omega_k := \sigma \tau_k, \quad (5)$$

so that the per-component survival function under the Gaussian-kernel HDP is lognormal: $1 - \Phi^N((\log t - \xi_{a,k})/\omega_k)$.

2 The survival difference Δ_{SD}

2.1 Definition

The survival (risk) difference at time t is

$$\Delta_{SD}(t; x, s_t) := S_1(t \mid x, s_t) - S_0(t \mid x, s_t). \quad (6)$$

2.2 Closed form under the HDP

$$\Delta_{\text{SD}}(t; x, s_t) = \sum_{k=1}^K \pi_{s_t, k} \left[\Phi^{\text{N}} \left(\frac{\log t - \xi_{0,k}(x, s_t)}{\omega_k} \right) - \Phi^{\text{N}} \left(\frac{\log t - \xi_{1,k}(x, s_t)}{\omega_k} \right) \right]. \quad (7)$$

2.3 Interpretation

$\Delta_{\text{SD}}(t; x, s_t)$ is the absolute increase in the probability of surviving past time t attributable to assigning treatment versus control for individuals with covariates x in target population s_t . It lives on the probability scale, is bounded between -1 and 1 , and has a direct clinical reading: “treatment increases the chance of surviving to year t by Δ_{SD} percentage points.”

This is the absolute risk reduction (ARR), or its survival-scale analogue. It is the estimand most directly tied to clinical decision-making at a fixed horizon and is the natural quantity for computing the number-needed-to-treat $1/|\Delta_{\text{SD}}(t; x, s_t)|$ when the contrast is favourable (Cook and Sackett, 1995).

When to use. When a single clinically meaningful horizon exists — for example, “5-year survival in early-stage cancer” or “1-year survival post-transplant” — the SD is the most interpretable summary. It depends on the error distribution Φ_{s_t} through (7) and is therefore sensitive to misspecification of the tail when t extends beyond the RCT follow-up; in the present framework that sensitivity is absorbed by the HDP error.

Limitations. $\Delta_{\text{SD}}(t; x, s_t)$ is a pointwise contrast at a single t and can give a misleading picture when survival curves cross or when the gain is concentrated outside the chosen window. It does not summarise the full survival benefit. For these reasons, regulatory and HTA guidance recommends pairing the SD with an integrated summary such as the RMST (Royston and Parmar, 2013).

3 The restricted mean survival time difference Δ_{RMST}

3.1 Definition

The RMST under treatment a at horizon t^* is

$$\mu_a(t^*; x, s_t) := \int_0^{t^*} S_a(u \mid x, s_t) du, \quad (8)$$

and the RMST contrast is

$$\Delta_{\text{RMST}}(t^*; x, s_t) := \mu_1(t^*; x, s_t) - \mu_0(t^*; x, s_t). \quad (9)$$

3.2 Closed form under the HDP

The per-component lognormal RMST has the standard expression

$$\mu_a^{(k)}(t^*; x, s_t) = e^{\xi_{a,k} + \omega_k^2/2} \Phi^{\text{N}} \left(\frac{\log t^* - \xi_{a,k} - \omega_k^2}{\omega_k} \right) + t^* \left[1 - \Phi^{\text{N}} \left(\frac{\log t^* - \xi_{a,k}}{\omega_k} \right) \right], \quad (10)$$

and the contrast is the source-specific mixture

$$\Delta_{\text{RMST}}(t^*; x, s_t) = \sum_{k=1}^K \pi_{s_t, k} [\mu_1^{(k)}(t^*; x, s_t) - \mu_0^{(k)}(t^*; x, s_t)]. \quad (11)$$

3.3 Interpretation

$\Delta_{\text{RMST}}(t^*; x, s_t)$ is the expected gain in survival time within the window $[0, t^*]$ attributable to treatment, for individuals with covariates x in target population s_t . It lives on the *time scale*, in the same units as the outcome, and has a direct reading: “treatment yields Δ_{RMST} additional units of expected life within the first t^* units.”

The RMST has been advanced as a principal alternative to the hazard ratio, particularly when proportional hazards is implausible (Royston and Parmar, 2011, 2013; Uno et al., 2014). Its appeal is threefold: it has a transparent units-of-time interpretation; it is well-defined under non-proportional hazards including crossing survival curves; and it integrates the full survival benefit within the window rather than reporting a contrast at a single point. Uno et al. (2014) review its use in oncology trials and provide guidance on the choice of t^* .

When to use. The RMST is the natural headline estimand under the long-term-extrapolation goal of this work. Reporting $\Delta_{\text{RMST}}(t^*; x, s_t)$ at multiple horizons — e.g. at t^* inside the trial window, at t^* just past it, and at t^* far beyond it — traces how the survival benefit evolves with horizon and exposes the contribution of the HDP-tail to the inference.

Choice of t^* . The horizon t^* should be chosen ex ante and motivated by clinical relevance or by regulatory context. Royston and Parmar (2013) recommend t^* at or near the largest follow-up time at which a non-negligible proportion of the at-risk set remains under observation. In the present setting that recommendation is too conservative: the central methodological point is to extrapolate beyond the RCT follow-up, so t^* should deliberately exceed it.

Limitations. The RMST depends on t^* , and substantive conclusions can shift when t^* is varied. Beyond the OS follow-up, Δ_{RMST} inherits the full extrapolation uncertainty of the posterior and should be reported with appropriate credible intervals.

4 The acceleration factor AF

4.1 Definition

Under the AFT model (1), the contrast between treatment and control at fixed covariates x is a constant additive shift on the log-time scale. Equivalently, on the time scale it is a multiplicative *acceleration factor*. The causal acceleration factor at x is

$$\text{AF}(x) := \exp\{\eta_1(x, s_t) - \eta_0(x, s_t)\} = \exp\{\tau(x) + (1 - s_t)c(x)\}. \quad (12)$$

For the causal contrast in either population, we use $s_t = 1$ (where (1) is unconfounded by construction) and the confounding term vanishes:

$$\text{AF}_{\text{causal}}(x) = \exp\{\tau(x)\}. \quad (13)$$

By transportability of potential outcomes (Assumption 4 of the framework document), $\text{AF}_{\text{causal}}(x)$ is the same in the RCT and OS populations: it is a property of the causal mechanism, not of the source.

The corresponding observed contrast in the OS, $\text{AF}_{\text{obs,OS}}(x) = \exp\{\tau(x) + c(x)\}$, is the confounded analogue. Its discrepancy from $\text{AF}_{\text{causal}}(x)$ is precisely $\exp\{c(x)\}$ and quantifies the multiplicative bias on the time scale due to unmeasured confounding in the OS.

4.2 Closed form and HDP-independence

The acceleration factor is a direct functional of the BART forest for τ and does not involve the error distribution. In particular,

$$\text{AF}_{\text{causal}}(x) = \exp\{\tau(x)\} \quad (14)$$

is computable at each MCMC iteration as $\exp\{\tau^{(r)}(x)\}$, without reference to the cluster atoms or weights of the HDP. The full posterior of $\text{AF}_{\text{causal}}(x)$ is therefore available at the same cost as the posterior of $\tau(x)$.

4.3 Interpretation

$\text{AF}_{\text{causal}}(x)$ is the ratio by which treatment multiplies an individual's time scale: a unit with covariates x assigned treatment experiences event times that are $\text{AF}_{\text{causal}}(x)$ times those under control. The reading is symmetric: $\text{AF}_{\text{causal}}(x) = 1.5$ means treatment extends time-to-event by 50%; $\text{AF}_{\text{causal}}(x) = 0.8$ means treatment shortens it by 20%.

The acceleration-factor interpretation is the defining feature of AFT models and is articulated in detail in Wei (1992), Kalbfleisch and Prentice (2002), and Cox and Oakes (1984). Compared with the hazard ratio, the AF has the advantages that (i) it is directly interpretable on the time scale rather than on the hazard scale, (ii) it is meaningful even when hazards are non-proportional, and (iii) under AFT it equals the ratio of median (and any other quantile) survival times:

$$\frac{\text{Quantile}_q(T(1) | x)}{\text{Quantile}_q(T(0) | x)} = \text{AF}_{\text{causal}}(x) \quad \text{for all } q \in (0, 1), \quad (15)$$

a property sometimes called *quantile equivariance* (Wei, 1992; Hougaard, 1999).

When to use. The AF is the natural estimand whenever the goal is to summarise the treatment effect by *a single number per covariate value*, free of horizon, free of error-distribution assumptions, and on a transparent scale. In data-fusion settings, this is particularly convenient: $\text{AF}_{\text{causal}}(x)$ depends only on $\tau(x)$ and is therefore unaffected by the extrapolation uncertainty that enters the RMST and SD via the HDP tail. It can be reported alongside the SD and RMST as a horizon-free summary.

Limitations. The AF carries the full weight of the AFT structural assumption: that the treatment effect is a single time-scale acceleration that does not vary with t . When the true effect waxes or wanes over time — as for some immune-mediated therapies — a single $\text{AF}_{\text{causal}}(x)$ misrepresents the effect. In such cases, SD and RMST at multiple t or t^* are more informative.

A second caveat is that the AF is a multiplicative-on-time-scale measure and does not by itself communicate absolute benefit. A 50% acceleration on a baseline median of one month and on a

baseline median of ten years have very different clinical implications. Pairing the AF with the SD or RMST closes this gap.

5 Population-averaged versions

Each of the three estimands has a population-averaged counterpart obtained by marginalising over the covariate distribution of the target population:

$$\Delta_{\text{SD}}(t; s_t) = \mathbb{E}[\Delta_{\text{SD}}(t; X, s_t) \mid S = s_t], \quad (16)$$

$$\Delta_{\text{RMST}}(t^*; s_t) = \mathbb{E}[\Delta_{\text{RMST}}(t^*; X, s_t) \mid S = s_t], \quad (17)$$

$$\text{AF}_{\text{causal}} = \mathbb{E}[\text{AF}_{\text{causal}}(X)] \quad \text{or} \quad \exp(\mathbb{E}[\tau(X)]), \quad (18)$$

where the last definition distinguishes the population-averaged AF (which preserves the multiplicative interpretation pointwise but is no longer a ratio of population means) from the AF of the population-averaged log-time shift (which is). The two coincide only under degenerate $\tau(X)$. Pointwise reporting via the posterior projection summaries is generally preferable to a single marginal number unless heterogeneity is small.

The Monte Carlo estimator is in each case the empirical average over evaluation rows of the corresponding posterior draw, with Bayesian-bootstrap weights if covariate-distribution uncertainty is to be propagated.

6 Summary

- $\Delta_{\text{SD}}(t; x, s_t)$ — absolute survival-probability gain at fixed t . Probability scale; depends on HDP tail. Natural for fixed clinical horizons and number-needed-to-treat.
- $\Delta_{\text{RMST}}(t^*; x, s_t)$ — expected life-time gain within $[0, t^*]$. Time scale; depends on HDP tail. Natural headline estimand under the long-term extrapolation goal.
- $\text{AF}_{\text{causal}}(x) = \exp\{\tau(x)\}$ — multiplicative time-scale acceleration. Horizon-free; HDP-free; assumes the AFT shift structure. Natural complement summarising treatment heterogeneity in a single covariate-specific number.

References

- Cook, R. J. and Sackett, D. L. (1995). The number needed to treat: a clinically useful measure of treatment effect. *BMJ*, 310(6977):452–454.
- Cox, D. R. and Oakes, D. (1984). *Analysis of Survival Data*. Chapman & Hall, London.
- Hougaard, P. (1999). Fundamentals of survival data. *Biometrics*, 55(1):13–22.
- Kalbfleisch, J. D. and Prentice, R. L. (2002). *The Statistical Analysis of Failure Time Data*, 2nd ed. Wiley, New York.
- Royston, P. and Parmar, M. K. B. (2011). The use of restricted mean survival time to estimate the treatment effect in randomized clinical trials when the proportional hazards assumption is in doubt. *Statistics in Medicine*, 30(19):2409–2421.

- Royston, P. and Parmar, M. K. B. (2013). Restricted mean survival time: an alternative to the hazard ratio for the design and analysis of randomized trials with a time-to-event outcome. *BMC Medical Research Methodology*, 13:152.
- Uno, H., Claggett, B., Tian, L., Inoue, E., Gallo, P., Miyata, T., Schrag, D., Takeuchi, M., Uyama, Y., Zhao, L., Skali, H., Solomon, S., Jacobus, S., Hughes, M., Packer, M., and Wei, L. J. (2014). Moving beyond the hazard ratio in quantifying the between-group difference in survival analysis. *Journal of Clinical Oncology*, 32(22):2380–2385.
- Wei, L. J. (1992). The accelerated failure time model: a useful alternative to the Cox regression model in survival analysis. *Statistics in Medicine*, 11(14–15):1871–1879.